AgriGaurd smart irrigation & smart water management

Smarter water. Better irrigation. Sustainable farming.

Problem

Small-scale farmers face challenges such as:

1. Limited water availability
2. Inefficient irrigation practices
3. Over-irrigation and water wastage
4. Different soil types having different water requirements
5. Irrigation decisions often relying on fixed schedules
6. Limited access to affordable smart-farming technology

Question: How can we help farmers use limited water more efficiently while still providing crops with the water they need?

Solution: AgriGaurd

AgriGuard uses **real-time sensing + automated control + water management** to make irrigation decisions.

Zone-Based :
Each area can be monitored and irrigated independently.

Adaptive :
The system records drying rates and irrigation cycles.

Water-aware :
Available tank water is considered when allocating irrigation.

Weather-Assisted :
Weather information can improve irrigation decisions.

Safe :
Physical E-STOP and reset controls are incorporated.

Why can AgriGuard become a real product?

🛒 Marketability

- ✓ Vegetable growers
- ✓ Small-scale farmers
- ✓ Community gardens
- ✓ Small agricultural operations
- ✓ Water-conscious growers

🔧 Buildability

- ✓ The prototype uses commercially available components:
- ✓ ESP32
- ✓ Capacitive soil sensors
- ✓ Relay module
- ✓ Solenoid valves
- ✓ Water-level sensor
- ✓ LCD
- ✓ Standard switches and resistors

System & Technical Design

Sensing

- Soil sensors
- Water level

ESP32 MCU

- Moisture data
- Drying rate
- Weather data

Irrigation control

- Valve 1
- Valve 2
- Valve 3

Safety

- E-STOP button
- RESET button
- ON/OFF button

Bill of components

Components	Quantity	Estimated cost
ESP32 Dev board	1	R 220.00
Capacitive soil sensor	3	R 660.00
Ultrasonic sensor	1	R 22.00
Solenoid irrigation valve	3	R 276.00
4-channel relay module	1	R 50.00
LCD I2C board	1	R 103.00
ON/OFF latch switch	2	R 80.00
E-STOP latching switch	1	R 86.00
Pull up resistors PACK (2-4KΩ)	1	R 100.00
3D printed Encapsulator(Fillament)	1	R 200.00
Tubing/piping (5m)	1	R 130.00
Wiring connectors (5m)	3	R 78.00
Flow sensor	1	R 69.00
Total :		**R 2074.00**

AgriGaurd team lineup

What does this project need:
Alfred 🤗 👍

🎯 **Project Manager**
- ➢ Oversees project architecture and system integration
- ➢ Coordinates team activities and project milestones

🔬 **Project Researcher**
- ➢ Researches suitable sensors and electronic components
- ➢ Evaluates component performance, cost, and suitability

⚡ **Electronic & Mechanical Designer**
- ➢ Designs electronic circuits and system wiring
- ➢ Develops mechanical components and enclosure designs

💻 **Embedded Systems Engineer**
- ➢ Programs the ESP32 firmware
- ➢ Develops sensor-processing and irrigation algorithms

📑 **Project Documentation**
- ➢ Maintains technical documentation and project reports
- ➢ Documents system designs, testing, and development
- ➢ Prepares technical presentations and supporting materials